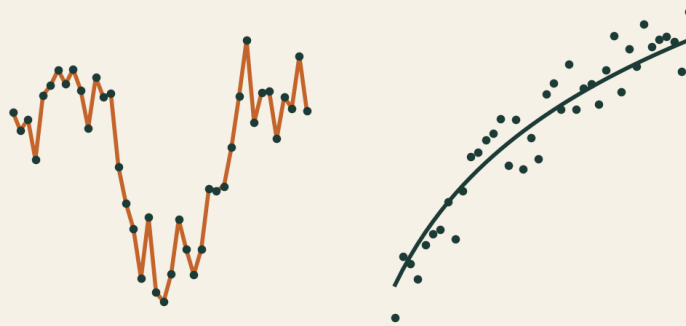


ROBUST REGRESSION ENGINE

Part H : Final Analysis & Reporting



Final Model Comparison & Business Recommendation Report

Best Model

Ridge / Lasso · R^2 0.9187

Regularization

Ridge vs Lasso · Stable Coefficients

Cross-Validation

4 Strategies · Within 1.3%

Supervised Learning

Part H: Final Analysis & Reporting

Consolidated findings from the Robust Regression Engine notebook: nine features, 3,800 properties, five model families, four cross-validation strategies. The results below use the actual metrics produced in the notebook.

1 Best-Performing Model and Justification

Ridge and Lasso Regression are the best-performing models, both reaching a validation R^2 of 0.9187, RMSE around ₹25.6 lakh, and MAE close to ₹19.6 lakh. Random Forest is a close second (R^2 0.9143), the pruned Decision Tree trails further behind (R^2 0.8835), and SVR fails to learn the relationship at all ($R^2 \approx -0.003$). The regularized linear models win because area, location score, bedrooms, bathrooms and property age relate to price in a way that is already close to linear, so a simple weighted-sum model captures nearly all of the signal without needing tree splits or kernels.

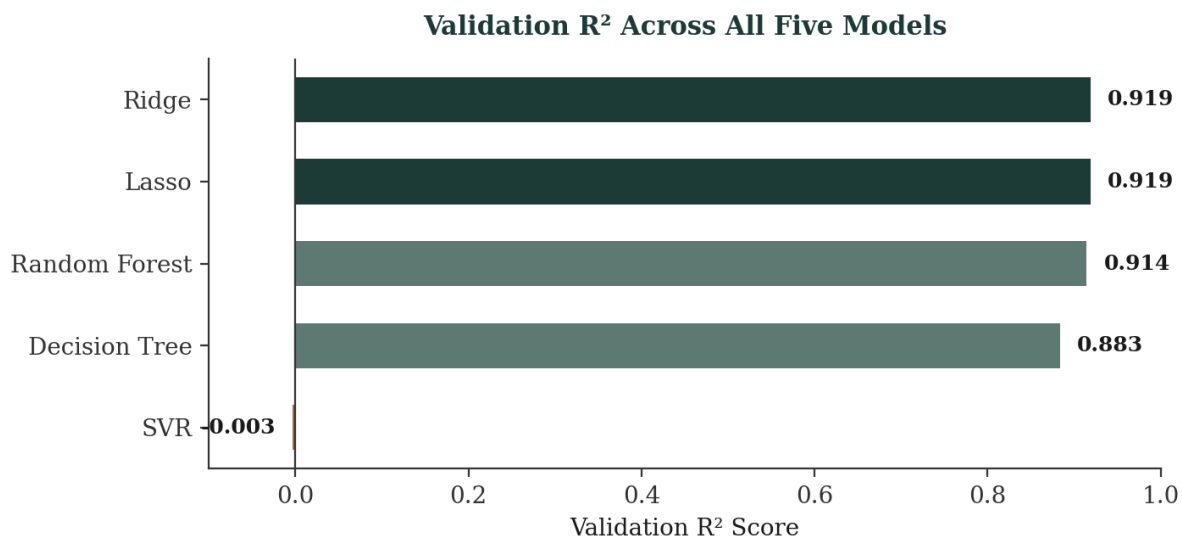


Figure 1 — Validation R^2 across all five model families; Ridge and Lasso lead, Random Forest is close behind, and SVR fails to generalize.

2 Impact of Regularization

Regularization was tuned with GridSearchCV: Ridge (L2) selected $\alpha = 1$, while Lasso (L1) selected $\alpha = 100$. Despite the very different penalty strengths, the resulting coefficients are almost identical — `area_sqft` and `location_score` remain the two strongest positive drivers, and `property_age` remains the strongest negative one. Lasso only mildly shrinks the weakest features (`distance_city_km`, `near_school`, `bedrooms`) and does not zero out any of them. This shows the nine features are all genuinely informative with little redundancy between them, so regularization here mainly stabilizes the model rather than performing feature selection.

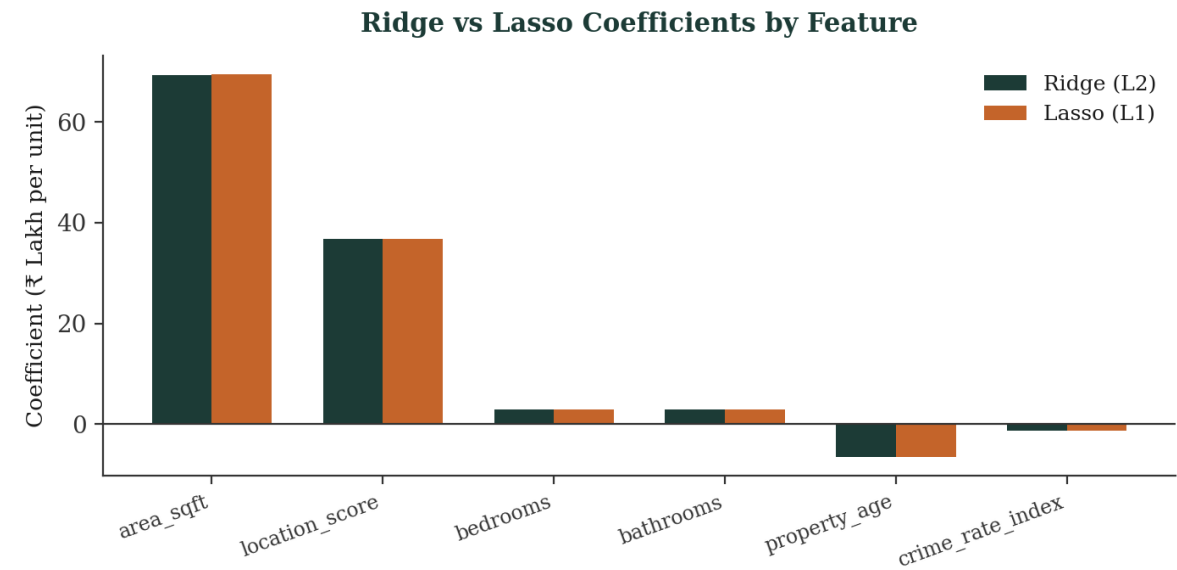


Figure 2 — Ridge (L2) and Lasso (L1) produce nearly identical coefficients per feature, confirming low multicollinearity in the data.

3

Role of Cross-Validation in Model Stability

Four cross-validation strategies were run on the Ridge model: K-Fold (avg. MSE 6.316×10^{12}), Stratified K-Fold on price quantiles (6.300×10^{12}), Leave-One-Out (6.298×10^{12}), and a chronological Time Series Split (6.379×10^{12}). All four estimates fall within about 1.3% of each other, so the single 80/20 test-set score is not a fluke — the model's performance is stable across different ways of slicing the data. Time Series Split gives the highest error, which is expected since it always validates on later, unseen time periods rather than a random mix of dates; it is also the most realistic estimate if this model will be used to price future sales.

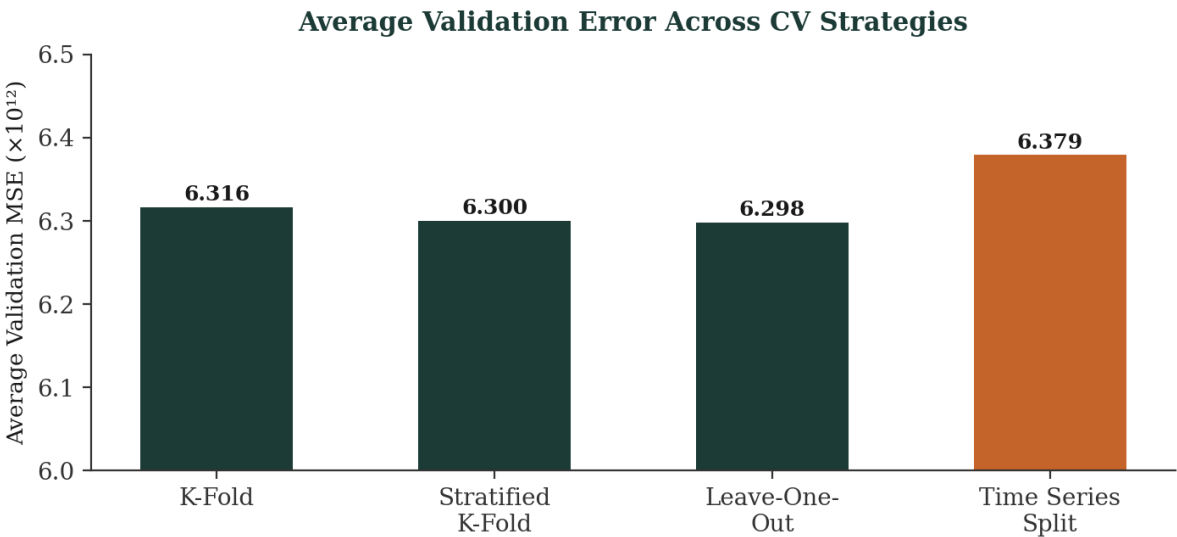


Figure 3 — Average validation error is nearly flat across K-Fold, Stratified K-Fold and LOOCV; Time Series Split is slightly higher due to its stricter, chronological validation.

4 Comparison of Linear vs Non-Linear Regressors

The regularized linear models (Ridge, Lasso) outperform every non-linear approach tested. Random Forest, a non-linear ensemble, comes close (R^2 0.9143) because averaging 100 trees smooths out the noise a single tree would overfit to. The pruned Decision Tree (R^2 0.8835) is a noticeably weaker single non-linear learner, and the unconstrained tree memorised the training data entirely (training $R^2 = 1.0$) before pruning fixed it. SVR, whether linear or RBF kernel, scored R^2 near zero in both cases — not because kernels are unsuitable, but because the default C and epsilon values are far too small relative to the multi-lakh price scale, and even the best grid-searched settings ($C=10$, $\gamma=0.1$, $\epsilon=0.1$) could not fix this within the tested range.

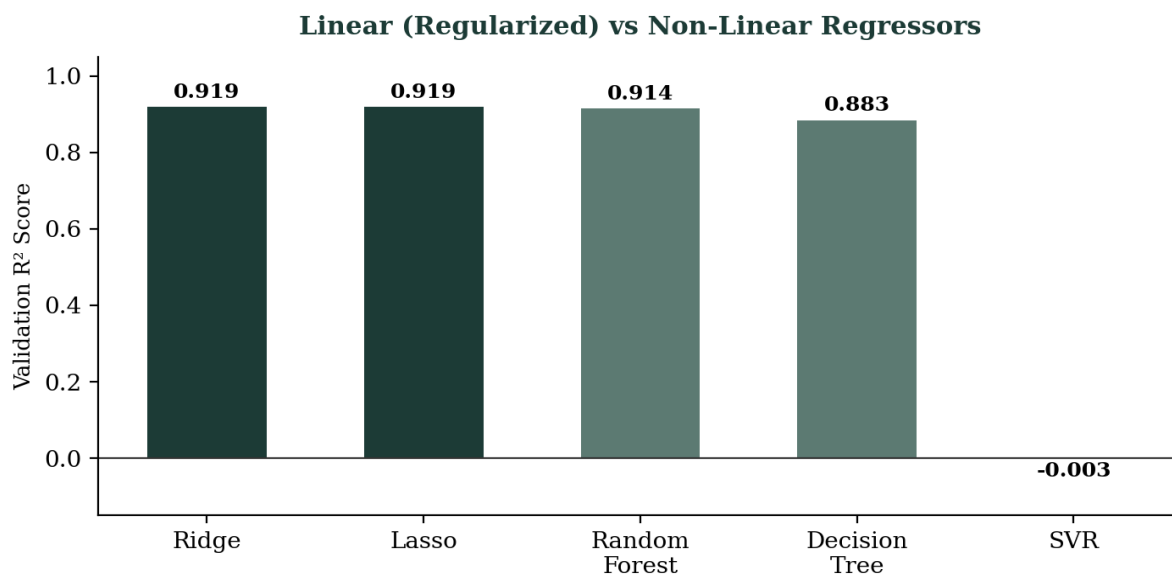


Figure 4 — Regularized linear models lead; Random Forest is a strong non-linear alternative; SVR underfits badly under its current configuration.

5 Business Interpretation of Results

For the business, three practical conclusions follow. First, Ridge or Lasso Regression should be the production model: it is the most accurate (R^2 0.9187), the cheapest to run, and its coefficients are directly explainable to non-technical stakeholders — each extra square foot or location-score point has a clear, quantified rupee impact. Second, `area_sqft` and `location_score` are the two levers that move price the most, while `property_age` and `crime_rate_index` pull price down, so valuation and renovation-investment decisions should weigh these first. Third, Random Forest is a reasonable backup where a slightly non-linear pattern is suspected, but SVR in its current form should not be deployed until the target is rescaled and retuned.

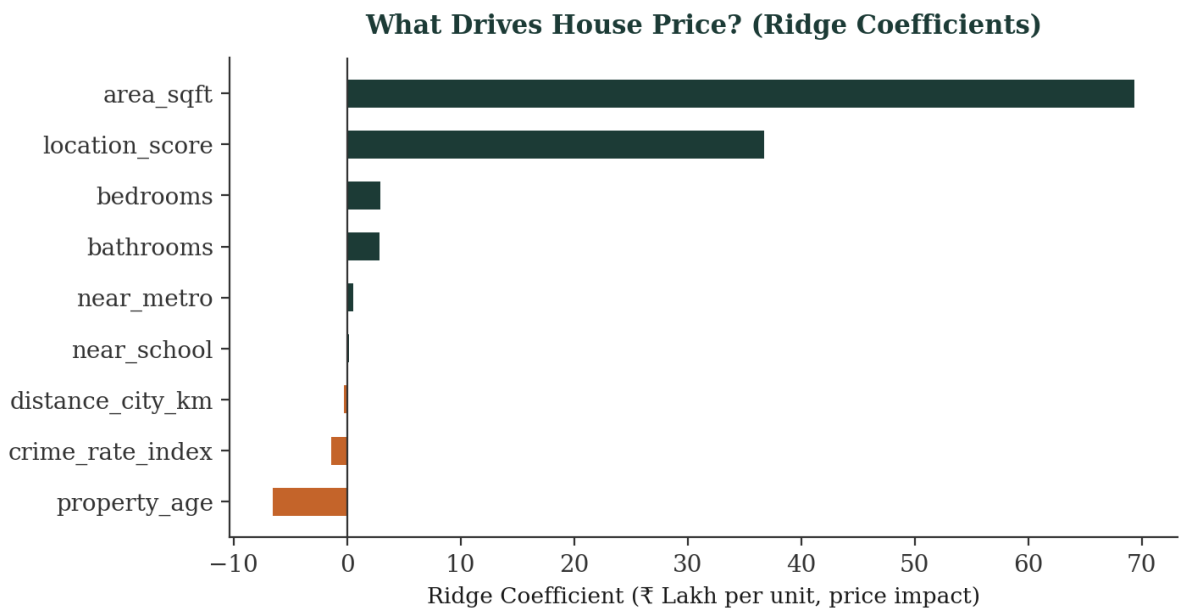


Figure 5 — Ridge coefficients ranked by rupee impact per unit; area and location score raise price the most, age and crime bring it down.

Question	Answer
Best model & why?	Ridge / Lasso — highest R ² (0.9187), lowest RMSE/MAE, fully interpretable coefficients.
Impact of regularization?	Ridge ($\alpha=1$) and Lasso ($\alpha=100$) converge to nearly the same coefficients — the data has little redundancy to shrink away.
Role of cross-validation?	K-Fold, Stratified, LOOCV and Time Series Split all agree within 1.3%, confirming the result is stable, not a lucky split.
Linear vs non-linear?	Linear models win; Random Forest is a close non-linear alternative; SVR underfits without target scaling.
Business takeaway?	Use Ridge/Lasso for pricing; area and location score are the biggest levers; keep Random Forest as backup.